

Few target observations improve river DOC reconstruction: regional prediction with station residual calibration

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Abstract

Reconstructing river dissolved organic carbon (DOC) at poorly monitored stations requires transferring information across catchments while accounting for differences in local concentration levels. We develop a two-stage reconstruction approach that combines a regional tree ensemble with a small target-station residual correction. The regional model learns from 40 environmentally similar source stations; one, three, or five target observations then calibrate its predictions without refitting the model. On 43 held-out stations in a 357-station river-network cohort, five observations per station reduce mean absolute error from 2.466 to 2.023 mg/L, a 17.93% decrease on the same 2,316 query cells. The paired, station-resampled interval for the cell-weighted difference is $[-1.026, -0.089]$ mg/L. The correction improves 36 of 43 stations, with substantial variation in benefit. The monotonic support curve and a station-shuffle diagnostic show the value of matching local corrections to their target stations; alternative tree learners give no further improvement. The study provides a practical way to combine regional prediction with sparse local measurements for historical water-quality reconstruction.

Keywords: dissolved organic carbon; sparse monitoring; spatial transfer; residual calibration; river water quality

1 Introduction

River water chemistry connects watershed conditions with downstream ecological environments. Yet monitoring records are uneven across stations, constituents, and time. The CAMELS-Chem data compilation illustrates both the opportunities created by combining hydroclimatic and chemistry records and the changing sampling coverage of individual solutes [1]. Reconstructing these gaps can make historical records more useful for comparing catchments and examining water-quality variability. This requires particular care at stations whose DOC observations were absent from model fitting.

Regionalization transfers information from observed catchments to less observed locations using environmental similarity or relationships between catchment attributes and hydrological response [2]. Learning methods offer complementary representations: tree ensembles fit nonlinear predictor-response relationships [3, 4], while recurrent models can combine temporal inputs with catchment attributes [5]. These approaches provide a regional starting point. They still leave an operational question: how should a model use a handful of target-station measurements once those measurements become available?

We address that question by separating regional prediction from station-level calibration. A model fitted on environmentally similar source stations supplies an initial DOC reconstruction. A shrunken mean residual, calculated from a small support set at the target station, adjusts its

concentration level in log-transformed space. The same regional predictions and query cells are retained at every support budget. The comparison therefore isolates what is gained by adding local measurements to a fixed regional model.

The contribution is a transparent reconstruction strategy and an empirical decomposition of its performance: a nested support curve on previously withheld stations, a comparison with alternative tree learners, and station-level diagnostics of where calibration helps or fails. The study tests sparse local adaptation within a known river-network cohort; the temporal graph model remains a separate component of the wider project.

2 Materials and methods

2.1 Cohort and evaluation population

The ST357 cohort contains 357 stations, 324 directed river-network edges, and 654 monthly positions, from April 1972 to September 2026. The current experiment concerns DOC concentration, expressed in mg/L. The station-holdout split contains 43 target stations with 2,531 observed station-month cells. Five observed months per target station are reserved as support candidates, leaving 2,316 query cells. All reported support levels use this identical query set. These counts describe observed cells available for evaluation, not the complete station-by-month reconstruction grid.

Model fitting excludes target-station DOC labels. The setting allows hydroclimatic measurements, ecological descriptors, and network connectivity at the target stations to be known. The station-holdout design addresses a different prediction problem from randomly masking individual records, a distinction emphasized by work on cross-validation for structured ecological data [6]. This is one outer station split within ST357, rather than a separate external basin.

2.2 Regional source selection and prediction

For each station we construct a descriptor from temperature and discharge summaries (mean, standard deviation, 10th, 50th, and 90th percentiles, and observation fractions), coordinates, ecological regime variables, graph degrees, and sums of incoming edge attributes. Descriptors are standardized across the known station cohort without reading DOC labels. The 40 source stations with the smallest squared Euclidean descriptor distance supply fitting records for each target-specific regional model. Source-pool size was selected using an internal station holdout.

The retained source model is an ExtraTrees regressor [3] with 120 trees, minimum leaf size four, all predictors eligible at each split, and seeds 42–46. Predictors include temperature and discharge with visibility flags, month sine/cosine terms, coordinates, ecological variables, visible same-month DOC context summaries, and target-history lag channels. The context summaries include global, upstream, and downstream visible values and support fractions. Thus, restricting the fitting pool to 40 stations does not restrict all contextual input to those 40 stations. This regional model uses explicit network summaries and should not be described as purely local or graph-free.

Training features use source training and designated context labels; final inference may additionally use source validation labels under the existing prediction policy. Target-station DOC remains hidden when the base predictions are generated. Source selection uses descriptor statistics across the historical record, making this a reconstruction setting with available covariates rather than a test of forward prediction from information available at a single historical date.

2.3 Nested target support and residual calibration

For each target station, the support candidates are selected from its sorted observed months without inspecting DOC values. The ordered candidates are the first, middle, last, first-quartile, and third-quartile positions, using the integer indices of an equally spaced five-position grid. The first K candidates define the nested support set $\mathcal{S}_i^{(K)}$, for $K \in \{0, 1, 3, 5\}$. All five candidates remain excluded from evaluation even when fewer are opened. Since support can occur after some query months, the task is retrospective reconstruction.

Let $z_{i,t} = \log(1 + y_{i,t})$ and let $\hat{z}_{i,t}^{\text{reg}}$ be the fixed regional prediction. For $K > 0$, the station residual and calibrated prediction are

$$\bar{r}_i^{(K)} = \frac{1}{K} \sum_{s \in \mathcal{S}_i^{(K)}} [z_{i,s} - \hat{z}_{i,s}^{\text{reg}}], \quad (1)$$

$$\hat{y}_{i,t}^{(K)} = \max \left\{ 0, \exp \left(\hat{z}_{i,t}^{\text{reg}} + \alpha_K \bar{r}_i^{(K)} \right) - 1 \right\}. \quad (2)$$

For $K = 0$ the correction is zero. The coefficients $(\alpha_1, \alpha_3, \alpha_5) = (0.25, 0.50, 0.75)$ are inherited unchanged from the earlier 300-tree internal station-holdout adapter analysis, which considered $\alpha \in \{0, 0.25, 0.50, 0.75, 1\}$. They are applied here to the frozen 120-tree product without a new parameter search. The base model is not refitted, and support labels enter only this residual calculation. In concentration space, the correction scales $1 + \hat{y}^{\text{reg}}$ by $\exp(\alpha_K \bar{r}_i^{(K)})$ before applying the lower bound. It adjusts concentration level, not the timing of events.

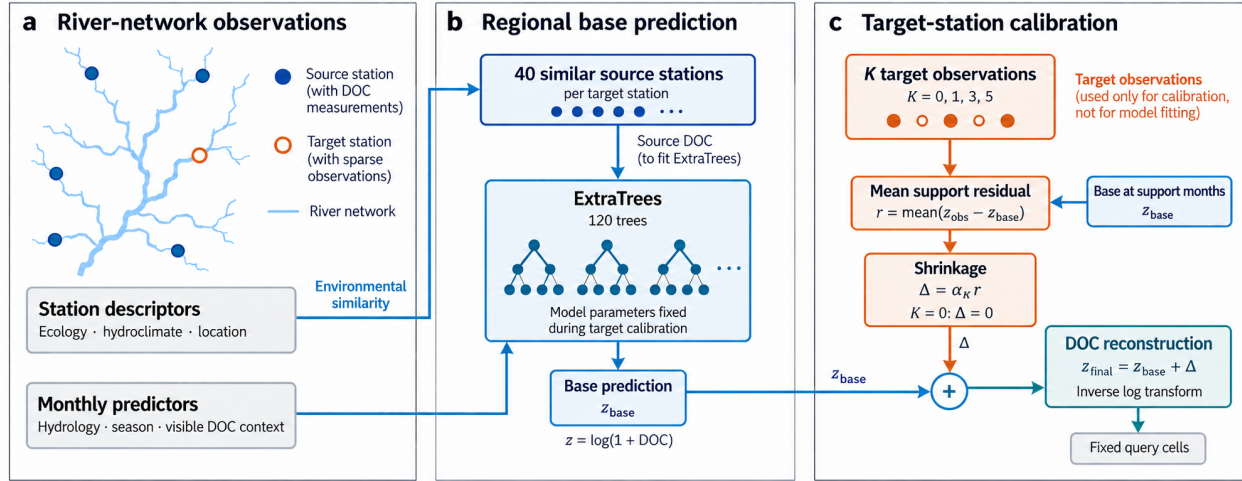


Figure 1: Regional prediction with sparse station residual calibration. **a**, Station descriptors select environmentally similar sources; monthly predictors include visible DOC context from the river network. **b**, A target-specific ExtraTrees model is fitted on 40 source stations and then held fixed. **c**, Target observations estimate a shrunk mean residual in $\log(1 + \text{DOC})$ space. This correction is added to the base prediction before inverse transformation and nonnegative clipping. At $K = 0$ the correction is zero. Support labels enter only calibration; query labels enter only evaluation. River branches and tree icons are schematic.

2.4 Comparisons and statistical summaries

The primary loss is MAE over the 2,316 query cells, averaged across five training seeds. We average losses, rather than calculating the error of an averaged prediction. RMSE is reported alongside

it; log-space MAE is retained in the accompanying prediction product. Station-level MAE weights query months equally within a station, and the station-macro summary then weights stations equally. These are distinct summaries when record lengths differ.

For paired confidence intervals, we first average losses across seeds at each cell, then resample 43 whole stations with replacement. Every sampled station contributes all its query cells, including multiplicity, and each replicate recomputes the cell-weighted MAEs and their difference. We use 5,000 resamples. Seed dispersion is reported separately; repeated seeds do not add independent stations.

A shuffle diagnostic permutes the fitted $K = 5$ station corrections across the 43 target stations, retaining the original query predictions. We use one predetermined permutation per seed at each support level (five in total per level) and average their MAEs. This is a descriptive mismatch diagnostic, not a permutation significance test. Station bias versus improvement is a descriptive analysis: both quantities use query errors and therefore do not provide an independent test of an ecological mechanism.

We also inspect a three-seed comparison of ExtraTrees, random forest, and histogram gradient boosting under the same regional selection protocol. Internal validation chose random forest in that later learner comparison. Its outer result is reported as selected; all three scores are shown to document the outcome. The outer split has been examined during multiple development rounds, so these comparisons are development evidence rather than a fresh confirmatory or external test.

3 Results

3.1 Five target observations reduce DOC reconstruction error

The primary MAE decreases from 2.466 mg/L without target support to 2.320, 2.160, and 2.023 mg/L at $K = 1, 3$, and 5, respectively (Table 1). Five observations per station reduce error by 17.93%. The matching cell-weighted paired difference is -0.442 mg/L, with a station-resampled 95% interval of $[-1.026, -0.089]$ mg/L. The station-equal difference is -0.419 mg/L; it is a complementary summary rather than the point estimate underlying the main table.

Table 1: Reconstruction on a fixed query. Errors average per-seed losses over the same 2,316 cells. SD describes five-seed variation; the intervals in the text resample stations.

K	MAE (mg/L)	Seed SD	RMSE (mg/L)	Reduction (%)
0	2.466	0.007	5.390	0.00
1	2.320	0.013	4.968	5.90
3	2.160	0.011	4.746	12.41
5	2.023	0.008	4.412	17.93

3.2 Benefit varies across target stations

Calibration improves 36 of 43 target stations. The median station-level relative reduction is 3.39%, compared with 17.93% for the pooled query. The two summaries differ because a reduction pooled over all cells gives greater weight to longer records and larger absolute errors. The map in Figure 3 displays both improvements and deteriorations from the same five-seed product as the main table.

Larger absolute baseline bias is associated with a larger absolute MAE reduction (Pearson $r = 0.92$; Figure 4). This is consistent with the calibration correcting station-level offsets, but the shared baseline in both axes contributes to their association. The correlation falls to 0.57 when the

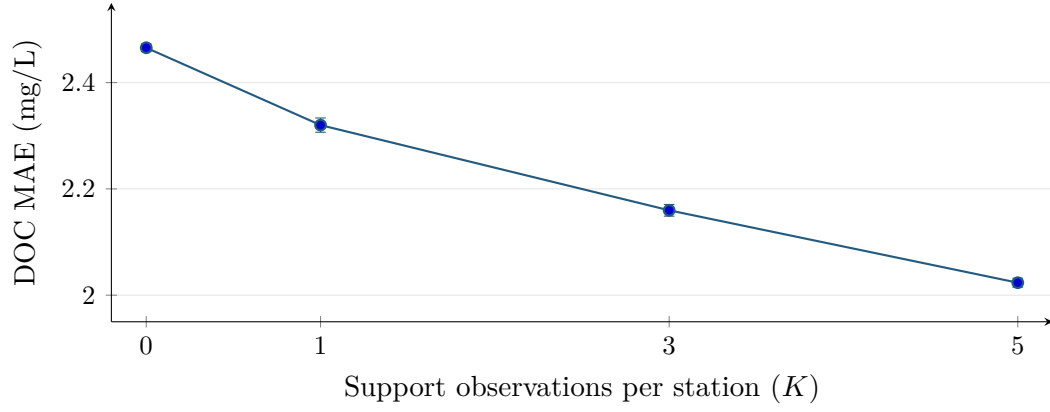


Figure 2: The support curve uses a fixed regional model and a fixed query. Error bars show one standard deviation of MAE across five seeds, not a confidence interval over stations.

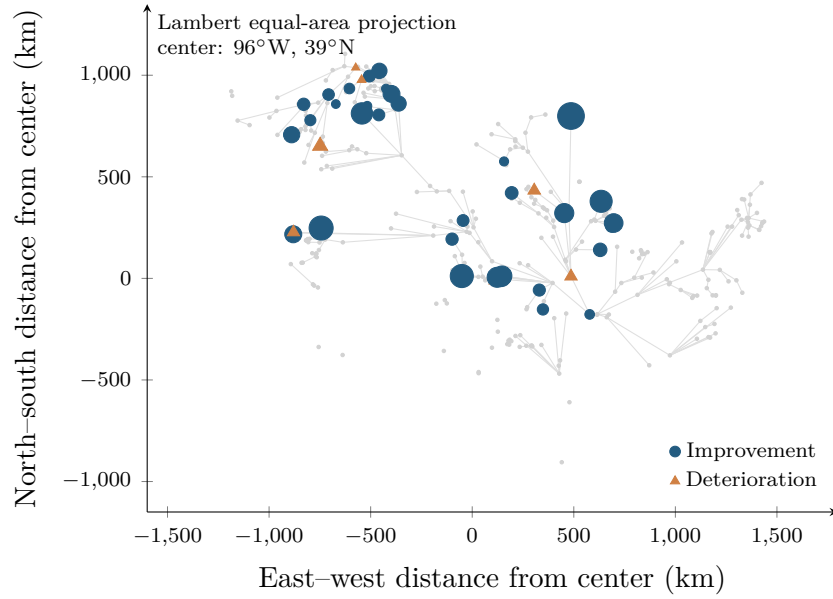


Figure 3: Station-level response to five support observations. Blue circles indicate lower MAE, and orange triangles indicate higher MAE; size scales with the absolute percentage change. Grey points show the station cohort, with straight graph links for geographic context rather than mapped river reaches.

station with the largest baseline bias is omitted; the rank correlation across all stations is 0.49. The descriptive shuffle diagnostic directly perturbs that matching: with station corrections reassigned, mean $K = 5$ MAE is 2.736 mg/L, compared with 2.023 mg/L for the matched corrections. These permutations describe sensitivity to station matching, not a further independent test set.

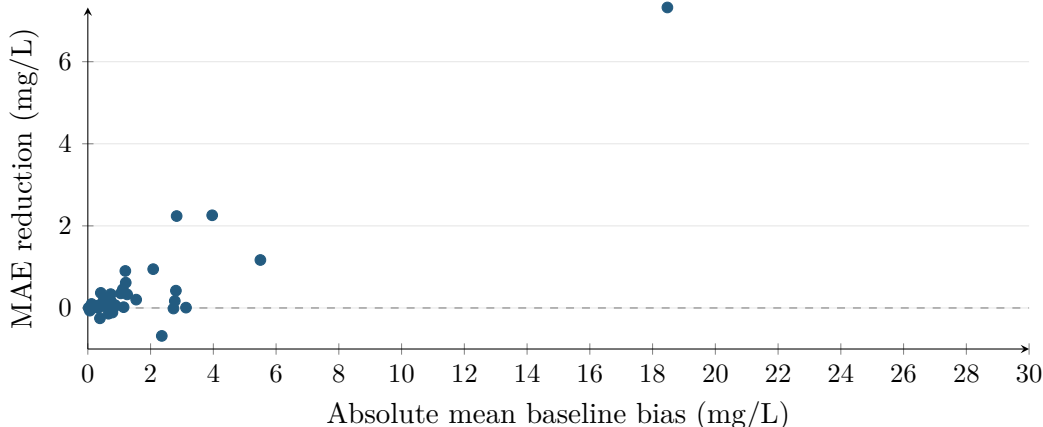


Figure 4: Descriptive relationship between absolute baseline bias and calibration gain. Each point is a target station, using seed-averaged errors. Both axes contain the baseline error, so the association should not be interpreted as independent causal evidence.

3.3 Alternative learners do not improve the regional result

The three-seed regional comparison gives $K = 5$ MAEs of 2.024 mg/L for ExtraTrees, 2.048 for random forest, and 2.086 for histogram gradient boosting. Random forest, selected by the internal split, therefore does not outperform the existing ExtraTrees reconstruction on the outer query. Gradient boosting gives identical predictions across its three seeds in this configuration; those repeats add no new stochastic evidence. The five-seed ExtraTrees value of 2.023 mg/L remains the reference result.

The main comparison isolates adaptation within the regional model. A matched quantitative comparison with the project’s local–transport neural model remains a separate analysis; numerical summaries not traceable to their prediction products are excluded from this manuscript.

4 Discussion

4.1 Local measurements complement regional prediction

The experiment shows that a small number of target DOC measurements can improve a regional reconstruction without fitting new model weights. Its useful operation is simple: compare observed and predicted concentrations at the same support months, then apply a shrunk correction to the remaining record. Subtracting the base prediction at each support month helps distinguish a station offset from the time-specific base prediction.

This decomposition gives regional models and local monitoring complementary roles. Environmental similarity supplies the initial model, while target measurements correct part of the station mismatch. The result aligns with the broader use of catchment attributes to represent heterogeneous hydrological behavior [5], although this study has not isolated a causal role for individual

ecological or hydrological predictors. A constant correction also cannot repair event timing, changing sources, or time-varying concentration–discharge relationships; those remain candidates for further model improvement.

4.2 Implications for spatiotemporal graph modeling

The broader project uses an ecological encoder, time memory, and directed river messages in a local–transport neural model. The present spatial experiment suggests a concrete target for that model: predicting the residual pattern that remains after regional context and sparse station calibration. Explicit network context is already present in the regional tree inputs. Demonstrating an additional graph benefit will therefore require comparing matched inputs and identifying which errors the learned message pathway resolves. The current spatial result does not by itself establish a transport mechanism.

4.3 Limits of transfer and monitoring interpretation

The study evaluates DOC on a single station holdout within the known ST357 cohort. Five support observations are a per-station budget (215 observations in total), selected across each station’s historical record. The result therefore concerns retrospective reconstruction after those observations are available, not a forecast before they are collected. Station descriptors also use the available historical covariates. Inference at a genuinely new basin or under prospective data availability needs a separate evaluation.

The station bootstrap reflects heterogeneity within this chosen holdout; it neither accounts for all dependence between connected stations nor removes the effect of repeated model development against the same outer split. Replication across independent station groups, support schedules, and basins is the next empirical requirement for a broader transfer claim. Finally, the wide range of station responses means that a pooled gain alone is insufficient for deciding where extra sampling is most valuable. The current work has not established an active-monitoring policy.

5 Conclusion

Regional prediction combined with sparse station calibration improves historical DOC reconstruction at held-out stations. On a fixed 2,316-cell query, five observations per station reduce MAE from 2.466 to 2.023 mg/L. The result identifies a practical source of predictive improvement: using local measurements to correct the regional concentration baseline. It also sets a clear target for further spatiotemporal graph development—explaining the variation that remains after this correction.

Data and code availability

Prediction products and analysis code are available in the local project repository; a public release has not yet been specified. The primary input is the frozen `regional_residual_product_v1` prediction table. The unified numerical and figure analysis is `manuscript_evidence_v2`, generated by `scripts/analyze_spatial_manuscript_evidence.py`. The three empirical plots contain numeric coordinates within the LaTeX source. The architecture schematic is supplied as a separate image asset alongside the source.

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